MIL-HDBK-1823 POD transmittal

Greetings:

Please tell me that you got this note because firewalls sometimes stop mail with attachments. Also, I'd like to know a bit about you and your work. What can you tell me? Thanks.

Since the software is mine, I am making it available to anyone who requests it, so here is the mh1823 POD software and test cases, etc. attached as a "ngzip" file. (I have no idea what "ngzip" means but apparently it will get past a firewall when ZIP won't.) Just rename it ZIP, unzip it and follow the directions on the Statistical Engineering website (https://statistical-engineering.com/mh1823/) and you should be OK. **Please read all the instructions before doing anything and you won't find things too onerous.** Especially don't unzip the ZIP file until you have created a directory for it, and then don't unzip the file within - R will do that when it installs the mh1823 POD package.

NOTE: The software has had many updates to improve functionality or to keep it compatible with R (that also is continually updated). You can see the history here: https://statistical-engineering.com/mh1823-history/

The mh1823POD software uses R, of course, and it also uses several R packages, most of which come with the basic R installation. You will need to install several required ancillary R packages. This is easy – within the R console, enter this command: `install.packages(c("tcltk2", "survival", "xlsx", "numDeriv"))`

Or you can install them individually: from R, choose Packages, Install package(s), and choose the package from the menu. If you don't have internet access, you can load the necessary R files locally. I have included them in this package.

***** Please let me know that you got this note. *****

I have received reports of difficulties installing the software and the most common cause is not having JAVA (both 32-bit and 64-bit versions) installed or not having the latest version. Java has been plagued recently with mischief and has issued frequent updates.

Background:
In 2007 I completed a year-long contract with the USAF to update the two-decades-old MIL-HDBK-1823 and the Air Force got around to approving the book in April 2009. I've included a copy with the mh1823 POD software.

The deliverables of my 2006-2007 contract to update MIL-HDBK-1823 included the draft of the updated Handbook and algorithms necessary to implement the methods described in the Handbook. Software was not a deliverable item. However, it soon became obvious that delivering descriptions of statistical algorithms, with no convenient implementation of them, would be of limited utility, so I wrote the mh1823 POD software suite to help promulgate the handbook's methods. I receive no financial support for this software apart from donations, so if you find the software useful and would like to contribute, you can do that here: https://statistical-engineering.com/support/
Thanks.

Caution:
I also want to caution you. While the methods in MIL-HDBK-1823A are effective for a very large majority of quantitative NDE situations, they are not universally applicable. There are at least two classes of data for which these methods are not valid:

1) Data that do not support a POD curve that goes to zero on the left, or to one on the right, i.e. POD "floor" and "ceiling" data (https://statistical-engineering.com/pod-floor-ceiling/), and

2) Field-Finds. (https://statistical-engineering.com/field-finds/) If you have these types of data, you will be able to torture the mh1823 POD software into giving you an answer. But it will be wrong.

Please note that I can teach a 2-day POD short course and workshop at your venue on the statistical foundations for the MIL-STD-1823A methods, and **using the advanced version of the mh1823POD software**. (https://statistical-engineering.com/mh1823-workshop/) *If you wish, we can use your enterprise data to solve real enterprise problems. And if the demands of a specific project are becoming onerous, you might consider me as a collaborator, especially if your task involves the interplay of fatigue/fracture, component life analysis, risk assessment and NDE efficacy.*

Best wishes!

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